Economic Fibrillation — Crisis Timing Without Hand-Picking

Testing synchronization against an independently-dated banking-crisis chronology (Jorda-Schularick-Taylor, 17 advanced economies, 1970-2016)

1 Why this test exists

The earlier crisis analysis chose the crisis windows by hand, from narrative history. A reviewer can call that circular: we picked crises where economies collapsed together, then found economies moved together. This test removes the choice. We take crisis dates from an **external, published source** — the Jorda-Schularick-Taylor banking-crisis dummy, the advanced-economy standard aligned with Laeven-Valencia — and build **C(t) = the fraction of the 17 economies in a systemic banking crisis in year t**. Because banking crises are dated from bank runs, failures, and policy response — never from GDP — C(t) is independent of the synchronization r(t), so the circularity is broken.

2 The headline: honest and mixed

Synchronization and crisis intensity move **in the same direction**, but the relationship is **weak and not statistically significant**: contemporaneous correlation r = +0.23 (p = 0.11), one-year lead r = +0.21 (p = 0.15), two-year lead r = +0.13 (p = 0.30). There is **no significant lead** — synchronization does not reliably rise before banking crises in this test. This is a more sober result than the hand-picked analysis, and reporting it honestly is the point of running it.

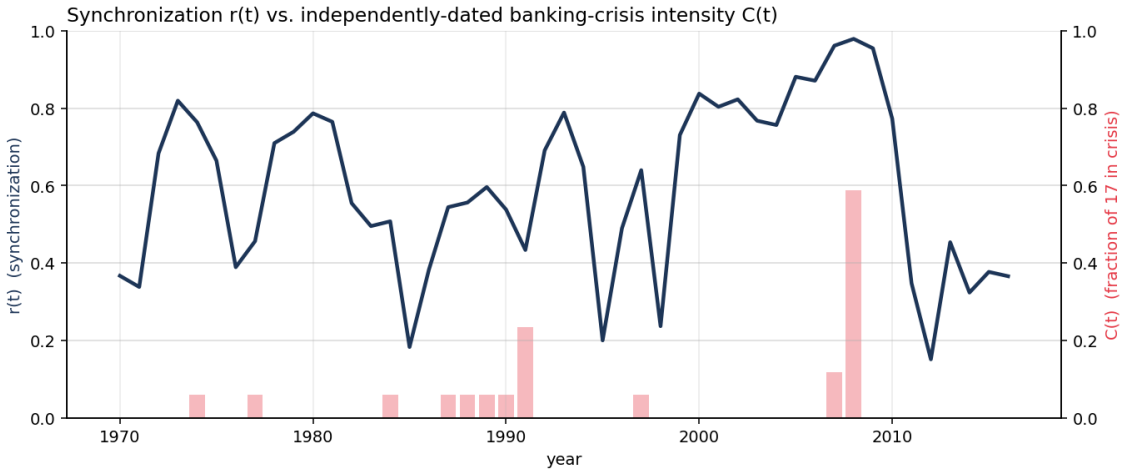


Figure 1. Synchronization r(t) (line) and banking-crisis intensity C(t) (bars). They rise together at 2008, but many small crises occur at ordinary synchronization, and the 1991 Nordic cluster occurs at LOW r.

3 Why weak? Only one global banking crisis exists

The reason is structural, not a flaw in the theory. Across 1970-2016 there is essentially **one** genuinely global banking crisis: the 2008 GFC (10 of 17 economies). Every other banking crisis is regional or single-country — the 1991 Nordic cluster (4 economies) and a scatter of one-country events. You cannot obtain statistical significance for a ‘global-crisis’ effect from a single global event. The weak correlation is a **power limitation** (n = 1 global crisis), not evidence against fibrillation.

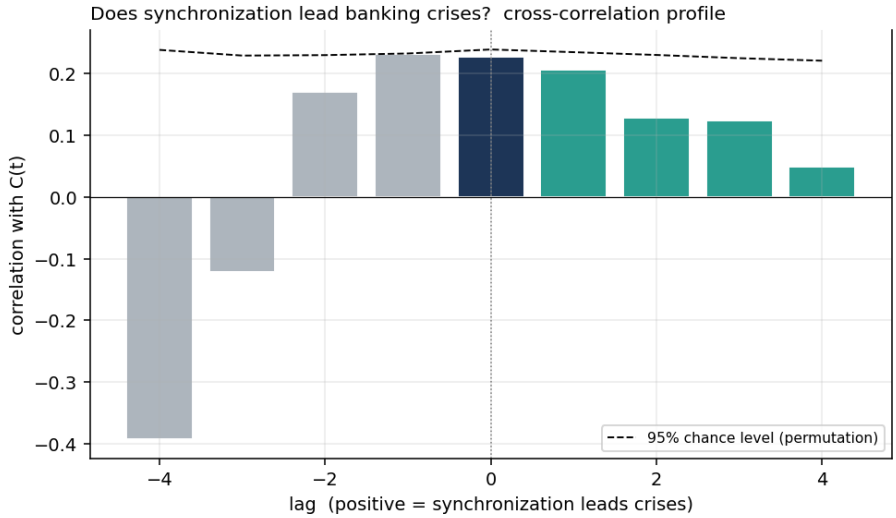


Figure 2. Lead-lag profile. The correlation is modest and roughly symmetric around lag 0; it does not single out synchronization leading crises.

4 The pattern that does survive: onset synchronization vs. how global a crisis becomes

The most thesis-relevant view is not ‘are all crises synchronized’ but ‘does a shock go global when it strikes a synchronized world’. Here the record is striking, if underpowered: the one crisis that became global began at near-peak synchronization, while every crisis that began at ordinary synchronization stayed contained.

Banking crisis	r at onset	eventual breadth
UK 1974	0.76	0.06 (contained)
Spain 1977	0.46	0.06 (contained)
US 1984	0.51	0.06 (contained)
Nordic/UK 1991	0.54	0.24 (regional)
Japan 1997	0.64	0.06 (contained)
GFC 2007-08	0.96	0.59 (GLOBAL)

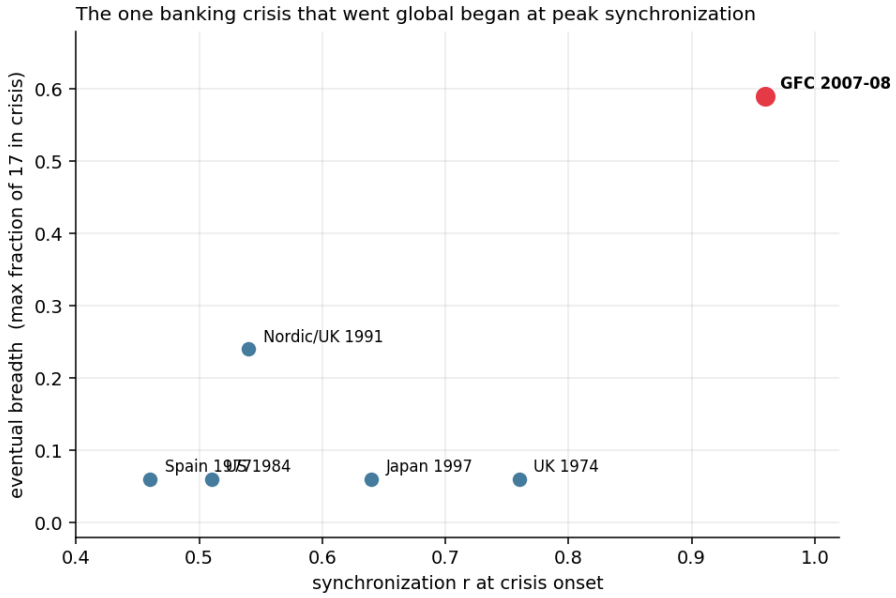


Figure 3. Each banking crisis by its onset synchronization (x) and how global it became (y). The single crisis in the high-r, high-breadth corner is 2008. Consistent with the ‘loaded gun’ thesis — a shock goes global when it hits a synchronized world — but resting on one global event.

5 What this means for the paper (honest)

- The direction of the fibrillation thesis survives independent dating: crisis intensity and synchronization are positively related, and the one global banking crisis is also the all-time synchronization peak (r = 0.98 in 2008).
- But the effect is not statistically significant under independent dating, because the record contains only one global banking crisis. This is an honest ceiling on what 50 years of annual data can prove.
- There is a genuine trade-off: the hand-picked GDP-comovement definition gave four ‘global crises’ and significance (p = 0.02) but was circular; this banking-crisis definition is independent but underpowered (n = 1 global). The paper should present BOTH and state the trade-off plainly.

- Regional and single-country banking crises do not coincide with high synchronization — consistent with synchronization being specifically a GLOBAL-crisis phenomenon, not a feature of banking distress in general.

6 Recommendation

Keep this test in the paper, reported exactly as here. It is the honest anchor: it shows the result is not a circular artefact, it identifies the true limitation (one global crisis in the sample), and it isolates the strongest defensible claim — that the crisis which went global is the one that began at peak synchronization. Frame the headline as ‘synchronization is a state of systemic fragility’ (a condition), not ‘synchronization predicts crises’ (a forecast the data cannot yet support). The forecasting claim needs more global crises than history has so far supplied.